Doira va aylana elementlari

Tanga, soat, g'ildirak — hammasi doira shaklida.

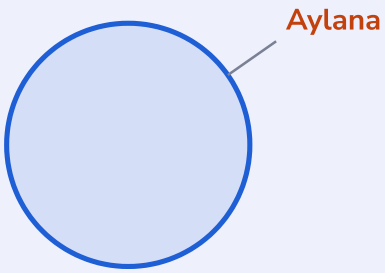
1-QISM

Mavzu

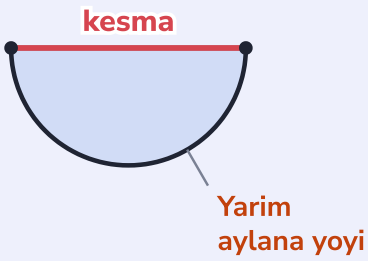


1-QOIDA · DOIRA, AYLANA, YARIM DOIRA, CHORAK DOIRA

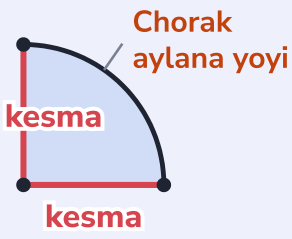
- Doiraning chegarasi **aylana** deb nomlangan egri chiziqdan iborat.
- **Yarim doira** — teng ikkiga bo'lingan doiraning bo'lagi. Yarim doira yarim aylana yoyi va kesma bilan chegaralangan.
- **Chorak doira** — teng to'rtga bo'lingan doiraning bir bo'lagi. Chorak doira chorak aylana yoyi va ikkita kesma bilan chegaralangan.



Doira



Yarim doira



Chorak doira

1-MISOL · BO'LINGAN TORT

Dumaloq tort markazidan o'tadigan ikkita kesik bilan 4 ta teng bo'lakka bo'lindi. Bir bo'lagi qanday shakl? U nimalar bilan chegaralangan?

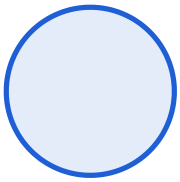
1. Doira teng to'rtga bo'lindi — bir bo'lagi **chorak doira**.
2. Uning chegarasi: tort chetidagi chorak aylana yoyi va markazdan chetgacha ketgan **ikkita kesma**.

2-MISOL · SHAKLNI TANIB OLISH

Qaysi shakl doira, qaysisi yarim doira, qaysisi chorak doira?



1 2



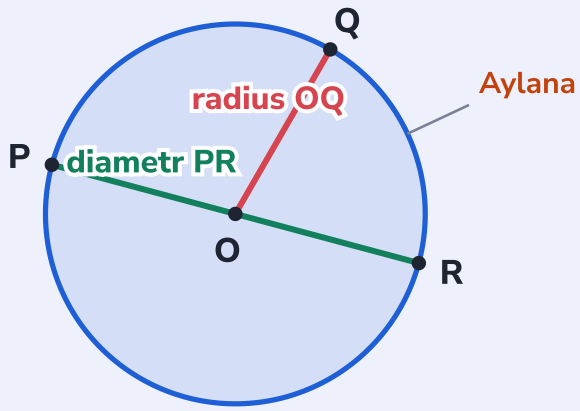
3

1. 1-shakl: chorak aylana yoyi va ikkita kesma — **chorak doira**.
2. 2-shakl: yarim aylana yoyi va bitta kesma — **yarim doira**.
3. 3-shakl: chegarasi butun aylana — **doira**.

2-QOIDA · MARKAZ, RADIUS, DIAMETR

- Aylananing ixtiyoriy nuqtasini uning markazi bilan tutashtiruvchi kesma **radius** deb ataladi.
- Aylana markazidan o'tadigan va aylanadagi ikkita nuqtani tutashtiruvchi kesma **diametr** deb ataladi.
- Doiraning markazi, radiusi va diametri mos ravishda uning aylanasining markazi, radiusi va diametri bo'ladi.
- Doiraning perimetri aylana yoyi uzunligiga teng bo'ladi.

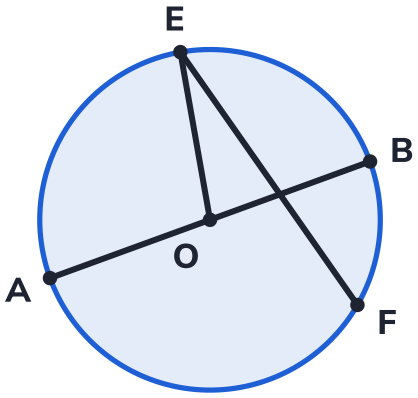
Eslatma: yarim doiraning kesmasi — doiraning **diametri**; chorak doiraning ikkala kesmasi — **radiuslar**.



O — markaz, OQ — radius, PR — diametr

1-MISOL · KESMALARNI AJRATISH

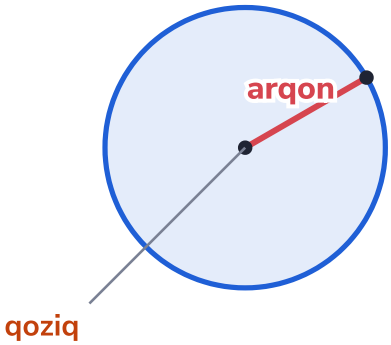
O — aylananing markazi. Rasmdagi kesmalardan qaysilari radius, qaysisi diametr? EF kesma diametr bo'ladimi?



1. Markazni aylanadagi nuqta bilan tutashtiruvchi kesmalar — radiuslar: **OA, OB, OE**.
2. AB markazdan o'tadi, uchlari aylanada — **diametr**.
3. EF ning uchlari aylanada, lekin u markazdan o'tmaydi — **diametr emas**.

2-MISOL · GULZORNI BELGILASH

Bog'bon yerga qoziq qoqdi va unga arqon bog'ladi. Arqonning ikkinchi uchini tarang tortib, qoziq atrofida aylanib chiqdi va yerda doira shaklidagi gulzorni belgiladi. Qoziq va arqon gulzorning qaysi elementlari?



1. Qoziq turgan nuqta — gulzorning **markazi**.
2. Arqon markazdan gulzor chetigacha yetadi — u **radius**.



Diametr ikkita radiusdan tuzilgan ekan-ku!

3-QOIDA · RADIUS VA DIAMETR BOG'LANISHI

- Doiraning barcha radiuslari teng uzunlikka ega.
- Doiraning diametri uning radiusidan ikki marta uzun.
- Doira radiusi diametrining yarmiga teng.

$$\text{Diametr} = 2 \cdot \text{radius}$$

$$\text{Radius} = \text{diametr} : 2$$

1-MISOL · RADIUS VA DIAMETR

a) Doiraning radiusi 9 cm. Diametrini toping. b) Doiraning diametri 17 m. Radiusini toping.

1. a) $\text{Diametr} = 2 \cdot 9 = 18$ cm.
2. b) $\text{Radius} = 17 : 2 = 8{,}5$ m.

2-MISOL · BOSHQA BIRLIKDA

Dumaloq stolning diametri 3 dm. Uning radiusi necha santimetr?

1. Avval santimetrga o'tkazamiz: 3 dm = 30 cm.
2. $\text{Radius} = 30 : 2 = 15$ cm.

2-QISM

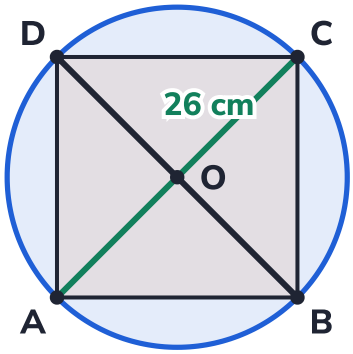
Misollar ishlash



Kesma markazdan o'tadimi? Hammasi shunga bog'liq.

1-MISOL · AYLANA ICHIDAGI KVADRAT

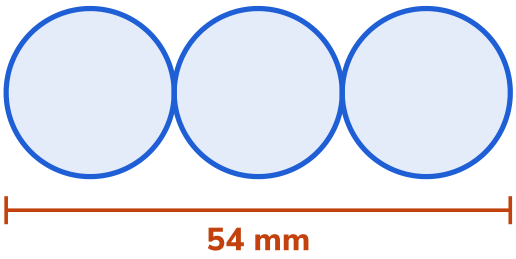
ABCD kvadratning hamma uchlari aylanada yotadi. Kvadratning diagonallari aylana markazi O da kesishadi. AC = 26 cm. Aylananing radiusini va BD kesma uzunligini toping.



1. AC markaz O dan o'tadi, uchlari aylanada — demak, AC **diametr**.
2. $\text{Radius} = 26 : 2 = 13$ cm.
3. BD ham markazdan o'tadi — u ham diametr. Barcha diametrlar teng: BD = 26 cm.

2-MISOL · TANGALAR QATORI

Uchta bir xil tanga stol ustiga bir-biriga tegib turadigan qilib qatorga terildi. Qatorning uzunligi 54 mm. Bitta tanganing radiusi qancha?



1. Qator uzunligi — uchta tanganing diametri.
2. Bitta diametr: $54 : 3 = 18$ mm.
3. $\text{Radius} = 18 : 2 = 9$ mm.

3-MISOL · QAYSI DOIRA KATTA?

Birinchi doiraning radiusi 5 cm, ikkinchisining diametri 9 cm. Qaysi doira katta?

1. Ikkalasining diametrini topamiz.
2. Birinchi doiraning diametri: $2 \cdot 5 = 10$ cm.
3. 10 cm > 9 cm — **birinchi doira katta** (5 < 9 bo'lsa ham!).



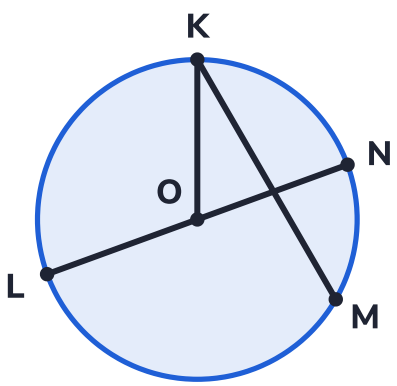
DIQQAT!

1. Aylanadagi ikki nuqtani tutashtiruvchi har bir kesma diametr emas. Diametr albatta markazdan o'tadi.
2. Radius → diametr: $\times 2$. Diametr → radius: $: 2$.
3. Doiralarni bir xil o'lchov bilan solishtiring.

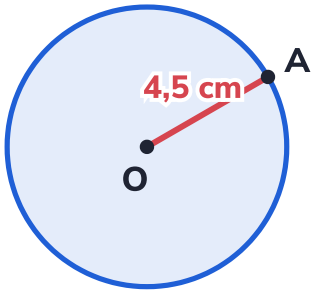
3-QISM

Mustahkamlash

1. O — aylana markazi. Kesmalarni ajrating: qaysisi radius, qaysisi diametr, qaysisi ikkalasi ham emas?



2. O — doira markazi, OA = 4,5 cm. Doiraning diametrini toping.

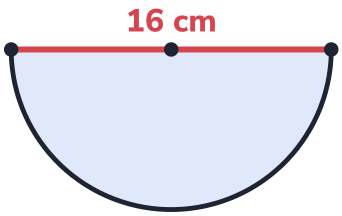


3. Doiraning diametri 19 dm. Radiusini toping.

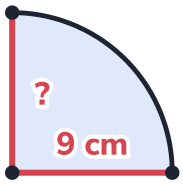
4. Jadvalni to'ldiring.

Doira	Radius	Diametr
a)	11 cm	?
b)	?	34 cm
c)	2,5 m	?
d)	?	9 mm

5. Yarim doiraning kesmasi 16 cm. Shu yarim doira radiusi qancha?



6. Chorak doiraning bir kesmasi 9 cm. a) Ikkinchi kesmasi qancha? b) Butun doiraning diametri qancha?



7. To'g'rimi yoki noto'g'rimi? a) Doiraning barcha diametrlari teng. b) Aylanadagi ikki nuqtani tutashtiruvchi har bir kesma — diametr. c) Radius diametrning yarmiga teng. d) Yarim doira ikkita kesma bilan chegaralangan.

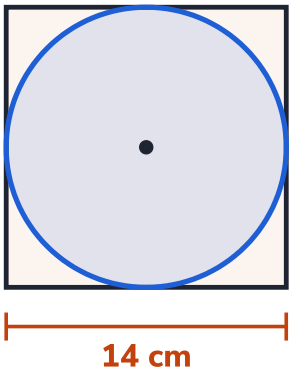
8. Devor soatining siferblati doira shaklida, diametri 28 cm. Uzun strelka markazdan siferblat chetigacha yetadi. Strelkaning uzunligi qancha?

4-QISM

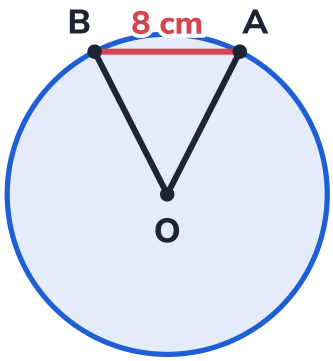
Mustaqil ish

1. Beshta bir xil tanga bir-biriga tegib turadigan qilib qatorga terildi. Qatorning uzunligi 1 dm. Bitta tanganing radiusi necha millimetr?

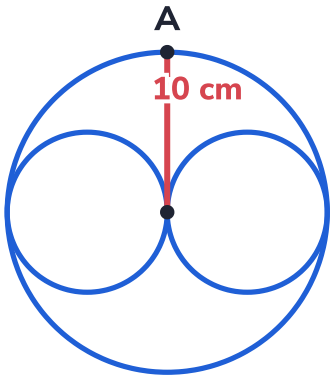
2. Kvadrat ichiga doira chizilgan, u kvadratning hamma tomoniga tegib turadi. Kvadratning tomoni 14 cm. Doiraning radiusini toping.



3. O — aylana markazi, A va B nuqtalar aylanada. AB = 8 cm, OAB uchburchakning perimetri 26 cm. Aylananing radiusini toping.

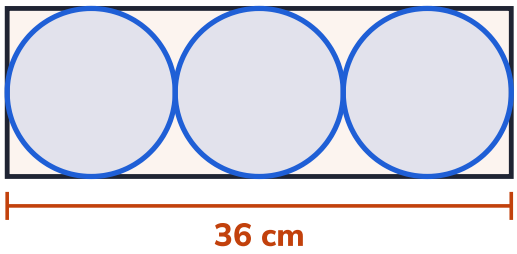


4. Katta doira ichida ikkita bir xil kichik doira bor. Ular katta doiraning diametri bo'ylab yonma-yon turadi va katta doiraga ichkaridan tegadi. Katta doiraning radiusi 10 cm. Kichik doiraning radiusi qancha?



5. **Xatoni toping.** Lola shunday yozdi: «Radiusi 8 cm bo'lgan doiraning diametri $8 : 2 = 4$ cm». Xato nimada? To'g'ri javob qancha?

6. **Qiyinroq.** To'g'ri to'rtburchak ichiga uchta bir xil doira qatorga zich joylashtirildi: ular bir-biriga va to'rtburchak tomonlariga tegib turadi. To'rtburchakning bo'yi 36 cm. a) Doiraning radiusi qancha? b) To'rtburchakning eni qancha? c) To'rtburchakning perimetri qancha?



5-QISM

Javoblar

Mustahkamlash

1-masala

OK — radius; LN — diametr (markazdan o'tadi); KM — ikkalasi ham emas (markazdan o'tmaydi).

2-masala

$\text{Diametr} = 2 \cdot 4{,}5 = 9$ cm.

3-masala

$\text{Radius} = 19 : 2 = 9{,}5$ dm.

4-masala

a) $2 \cdot 11 = 22$ cm; b) $34 : 2 = 17$ cm; c) $2 \cdot 2{,}5 = 5$ m; d) $9 : 2 = 4{,}5$ mm.

5-masala

Yarim doiraning kesmasi — butun doiraning diametri. $\text{Radius} = 16 : 2 = 8$ cm.

6-masala

a) Ikkala kesma ham radius — 9 cm. b) $\text{Diametr} = 2 \cdot 9 = 18$ cm.

7-masala

a) To'g'ri. b) Noto'g'ri — markazdan o'tmasa, diametr emas. c) To'g'ri. d) Noto'g'ri — bitta kesma va yarim aylana yoyi bilan chegaralangan.

8-masala

Strelka — radius: $28 : 2 = 14$ cm.

Mustaqil ish

1-masala

1 dm = 100 mm. Bitta diametr: $100 : 5 = 20$ mm. $\text{Radius} = 20 : 2 = 10$ mm.

2-masala

Doiraning diametri kvadrat tomoniga teng — 14 cm. $\text{Radius} = 14 : 2 = 7$ cm.

3-masala

OA = OB = radius. $OA + OB = 26 - 8 = 18$ cm. $\text{Radius} = 18 : 2 = 9$ cm.

4-masala

Ikki kichik doiraning diametrlari birgalikda katta doiraning diametriga teng: $2 \cdot 10 = 20$ cm. Bitta kichik doiraning diametri $20 : 2 = 10$ cm, radiusi $10 : 2 = 5$ cm.

5-masala

Lola radiusni 2 ga bo'ldi, aslida 2 ga ko'paytirish kerak edi: $\text{Diametr} = 2 \cdot 8 = 16$ cm.

6-masala

a) Bo'yi uchta diametrga teng, bitta diametr: $36 : 3 = 12$ cm. Radius: $12 : 2 = 6$ cm. b) Eni — bitta diametr: 12 cm. c) $P = 2 \cdot (36 + 12) = 96$ cm.